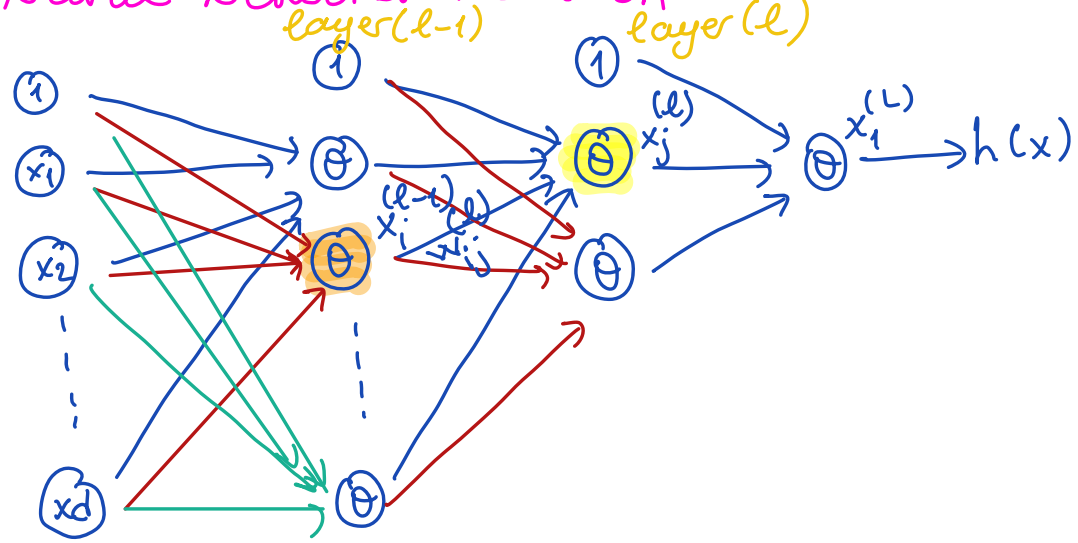
Neural Networks : Notation



input $\underline{x}$ layer 1 layer 2 layer 3 (output)

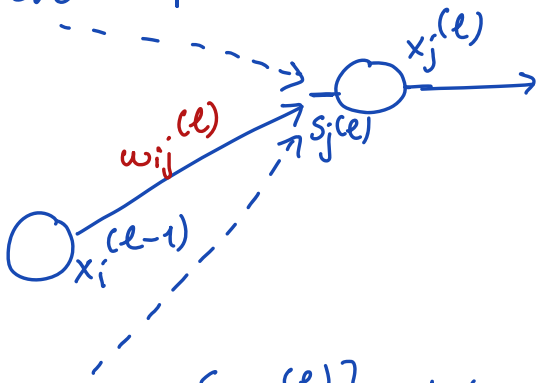
$$w_{ij}^{(l)} = \begin{cases} 1 \leq l \leq L & \text{layers} \\ 0 \leq i \leq d^{(l-1)} & \text{inputs} \quad (\#\text{ neurons at layer } l-1) \\ 1 \leq j \leq d^{(l)} & \text{output} \quad (\#\text{ neurons at layer } l) \end{cases}$$

$$x_j^{(l)} = \theta(s_j^{(l)}) = \theta\left(\underbrace{\sum_{i=0}^{d^{(l-1)}} w_{ij}^{(l)} x_i^{(l-1)}}_{s_j^{(l)}}\right) \quad \text{(recursive eq}^{n}\text{)}$$

Apply $\underline{x}$ to $\underbrace{x_1^{(0)} \cdots x_{d^{(0)}}^{(0)}}_{\text{inputs}} \longrightarrow \underbrace{x_1^{(L)} = h(\underline{x})}_{\text{output (a single number)}}$

When the weights $w_{ij}^{(l)}$ of the network are known, we can compute the output in a <u>feedforward</u> way as given by the recursive eq^{n} above.

Goal: Estimate the weights using learning so that the input and output are correct.



This is the only branch that $w_{ij}^{(l)}$ effects $e(\underline{w})$ (error)

All weights $\underline{w} = \{w_{ij}^{(l)}\}$ determine $h(\underline{x})$

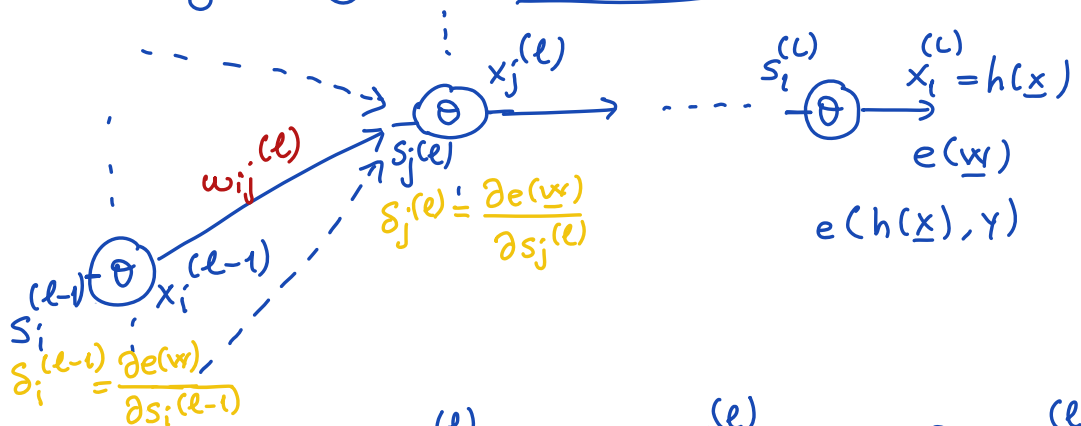
Error on example $(\underline{x}_n, y_n)$ is $e(h(\underline{x}_n), y_n) = e(\underline{w})$

Apply SGD by computing

$$\nabla e(\underline{w}): \quad \frac{\partial e(\underline{w})}{\partial w_{ij}^{(l)}} \qquad \forall\, i,j,l$$

We can evaluate this analytically or numerically.

Evaluate analytically using chain rule:



$$\frac{\partial e(\underline{w})}{\partial w_{ij}^{(l)}} = \underbrace{\frac{\partial e(\underline{w})}{\partial s_j^{(l)}}}_{\delta_j^{(l)}} \times \underbrace{\frac{\partial s_j^{(l)}}{\partial w_{ij}^{(l)}}}_{x_i^{(l-1)}}$$

: $w_{ij}^{(l)}$ only affects $s_j^{(l)}$

Final layer $\quad \delta_1^{(L)} = \dfrac{\partial e(\underline{w})}{\partial s_1^{(L)}}$

$$e(\underline{w}) = \left(x_1^{(L)} - y_n\right)^2 = \left(\theta(s_1^{(L)}) - y_n\right)^2$$

$$\theta'(s) = \left(1 - \theta^2(s)\right) \text{ for the tanh}$$

$$\delta_1^{(L)} = \frac{\partial e(\underline{w})}{\partial s_1^{(L)}} = \frac{\partial e(\underline{w})}{\partial x_1^{(L)}} \times \frac{\partial x_1^{(L)}}{\partial s_1^{(L)}}$$

$$= 2\left(x_1^{(L)} - y_n\right) \times \theta'\left(s_1^{(L)}\right)$$

$$\boxed{\delta_1^{(L)} = 2\left(x_1^{(L)} - y_n\right) \times \left(1 - \theta^2\left(s_1^{(L)}\right)\right)}$$

Backpropagation of δ :

$$\delta_i^{(\ell-1)} = \frac{\partial e(\underline{w})}{\partial s_i^{(\ell-1)}} = \sum_{j=1}^{d^{(\ell)}} \frac{\partial e(\underline{w})}{\partial s_j^{(\ell)}} \times \frac{\partial s_j^{(\ell)}}{\partial x_i^{(\ell-1)}} \times \frac{\partial x_i^{(\ell-1)}}{\partial s_i^{(\ell-1)}}$$

$$= \sum_{j=1}^{d^{(\ell)}} \underbrace{\delta_j^{(\ell)}}_{\text{known}} \times w_{ij}^{(\ell)} \times \theta'\left(s_i^{(\ell-1)}\right)$$

$$\boxed{\delta_i^{(\ell-1)} = \left(1 - \left(x_i^{(\ell-1)}\right)^2\right) \sum_{j=1}^{d^{(\ell)}} w_{ij}^{(\ell)} \delta_j^{(\ell)}}$$